

1-1

$$I_c = \log_2 \frac{1}{0.023} = 5.442 \text{ (bit)}$$

$$I_e = \log_2 \frac{1}{0.105} = 3.252 \text{ (bit)}$$

$$I_o = \log_2 \frac{1}{0.001} = 9.966 \text{ (bit)}$$

$$I_x = \log_2 \frac{1}{0.002} = 8.966 \text{ (bit)}$$

1-2.

$$\begin{aligned} H(X) &= \sum_{n=1}^N P(x_n) \log_2 P(x_n) \\ &= \frac{1}{2} + \frac{1}{4} \log_2 4 + \frac{1}{8} \log_2 8 + \frac{1}{8} \log_2 8 \\ &= 1.75 \text{ (bit/符号)} \end{aligned}$$

$$R_b = 1000 \times 1.75 = 1750 \text{ (bit/s)}$$

1-5: 像素 = 分辨率 × 尺寸

$$\text{像素} = 210 \times 297 \times \frac{100}{25.4} \times \frac{100}{25.4} = 9.67 \times 10^5$$

$$\text{每像素的平均信息量: } I_p = \log_2 256 = 8 \text{ bit}$$

$$\text{每幅图像的平均信息量: } I = 9.67 \times 10^5 \times 8 = 7.736 \times 10^6 \text{ bit}$$

$$\text{信噪比: } \frac{S}{N} = 10^2 = 100$$

$$\therefore C_{\max} = B_{\max} \log_2 (1 + \frac{S}{N}) = 3100 \times \log_2 (1 + 100) = 2.064 \times 10^4$$

$$\therefore t_{\min} = \frac{I}{C_{\max}} = \frac{7.736 \times 10^6}{2.064 \times 10^4} \approx 374.8 \text{ (s)}$$

1-6:

$$\text{每幅图像的总信息量: } I_1 = 640 \times 480 \times 24 = 7.38 \times 10^6 \text{ bit}$$

$$\text{每秒传输的总信息量: } I_2 = I_1 \times 25 = 1.84 \times 10^8 \text{ bit}$$

$$\therefore B = \frac{I_2}{\log_2 (1 + \frac{S}{N})} = \frac{1.84 \times 10^8}{\log_2 (1 + 100)} \approx 1.85 \times 10^7 \text{ Hz}$$

2-2.

$$\begin{aligned}
 f(t) &= A \cos(200\pi t) \cos(2000\pi t) \\
 &= \frac{A}{2} [\cos(200\pi t + 2000\pi t) + \cos(200\pi t - 2000\pi t)] \\
 &= \frac{A}{2} [\cos(2200\pi t) + \cos(1800\pi t)]
 \end{aligned}$$

信号的自相关函数:

$$\begin{aligned}
 R(\tau) &= \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) f(t+\tau) dt \\
 &= \frac{A^2}{4} \int_{-\frac{T}{2}}^{\frac{T}{2}} [\cos(2200\pi t) + \cos(1800\pi t)] [\cos(2200\pi(t+\tau)) + \cos(1800\pi(t+\tau))] dt \\
 &= \frac{A^2}{4} \left(\frac{1}{2} \cos(2200\pi\tau) + \frac{1}{2} \cos(1800\pi\tau) \right) \\
 &= \frac{A^2}{8} \cos(2200\pi\tau) + \frac{A^2}{8} \cos(1800\pi\tau)
 \end{aligned}$$

信号的功率谱密度:

$$\begin{aligned}
 P(f) &= \int_{-\infty}^{\infty} R(\tau) e^{-j2\pi f\tau} d\tau \\
 &= \int_{-\infty}^{\infty} \left[\frac{A^2}{8} \cos(2200\pi\tau) + \frac{A^2}{8} \cos(1800\pi\tau) \right] e^{-j2\pi f\tau} d\tau \\
 &= \frac{A^2}{16} [\delta(f+1100) + \delta(f-1100) + \delta(f+900) + \delta(f-900)]
 \end{aligned}$$

平均功率: $P = R(0) = \frac{A^2}{4}$

2-4. 设 $S(t) = m(t) \cos 2\pi f_c t$ 则 $S(t)$ 的希尔伯特变换为: $\hat{S}(t) = m(t) \cos 2\pi f_c t * \frac{1}{\pi t} = m(t) \sin 2\pi f_c t$ 解析信号: $\tilde{x}(t) = S(t) + j\hat{S}(t)$

$$\begin{aligned}
 &= m(t) \cos 2\pi f_c t + j m(t) \sin 2\pi f_c t \\
 &= m(t) (\cos 2\pi f_c t + j \sin 2\pi f_c t) \\
 &= m(t) e^{j2\pi f_c t}
 \end{aligned}$$

复包络: $f_L(t) = \tilde{x}(t) e^{-j2\pi f_c t}$

$$\begin{aligned}
 &= m(t) e^{j2\pi f_c t} \cdot e^{-j2\pi f_c t} \\
 &= m(t)
 \end{aligned}$$

2-10.

$$\begin{aligned}
 (1). E[\tilde{x}(t)] &= E[x_1 \cos 2\pi f_0 t - x_2 \sin 2\pi f_0 t] \\
 &= E(x_1) \cos 2\pi f_0 t - E(x_2) \sin 2\pi f_0 t \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 E[\tilde{x}^2(t)] &= E[(x_1 \cos 2\pi f_0 t - x_2 \sin 2\pi f_0 t)^2] \\
 &= E[x_1^2 \cos^2 2\pi f_0 t + x_2^2 \sin^2 2\pi f_0 t - 2x_1 x_2 \cos(2\pi f_0 t) \sin(2\pi f_0 t)] \\
 &= E[x_1^2 \cos^2 2\pi f_0 t + x_2^2 \sin^2 2\pi f_0 t] \\
 &= E\left[\frac{x_1^2}{2} (1 + \cos 4\pi f_0 t) + \frac{x_2^2}{2} (1 - \cos 4\pi f_0 t)\right] \\
 &= \frac{6^2}{2} + \frac{6^2}{2} \\
 &= 6^2
 \end{aligned}$$

①

(2). x 为一个均值为 0, 方差为 σ^2 的正态随机变量

$x(t)$ 的一维分布密度函数: $p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{x^2}{2\sigma^2}\right]$

(3). 自相关函数:

$$\begin{aligned} R_x(t_1, t_2) &= E[x(t_1)x(t_2)] \\ &= E[(x_1 \cos 2\pi f_0 t_1 - x_2 \sin 2\pi f_0 t_1)(x_1 \cos 2\pi f_0 t_2 - x_2 \sin 2\pi f_0 t_2)] \\ &= E[x_1^2 \cos(2\pi f_0 t_1) \cos(2\pi f_0 t_2) + x_2^2 \sin(2\pi f_0 t_1) \sin(2\pi f_0 t_2)] \\ &= \frac{1}{2} E(x_1^2) [\cos 2\pi f_0 (t_1 + t_2) + \cos 2\pi f_0 (t_2 - t_1)] + \frac{1}{2} E(x_2^2) [\cos 2\pi f_0 (t_2 - t_1) - \cos 2\pi f_0 (t_1 + t_2)] \end{aligned}$$

$$\because E(x_1^2) = E(x_2^2) = \sigma^2$$

$$\therefore R_x(t_1, t_2) = \sigma^2 \cos[2\pi f_0 (t_2 - t_1)]$$

协方差函数:

$$\begin{aligned} C_x(t_1, t_2) &= E[x(t_1)x(t_2)] - m_x(t_1)m_x(t_2) \\ &= R_x(t_1, t_2) \\ &= \sigma^2 \cos[2\pi f_0 (t_2 - t_1)] \end{aligned}$$

(4). $\because E[x(t)] = 0$ 为常数; 且自相关函数 $R_x(t_1, t_2)$ 只与时间间隔 $t_2 - t_1$ 有关

$\therefore x(t)$ 为宽平稳过程

$$2-12. (1) E[Y(t)] = 2 + 3E[X(t)] = 2 + 3 = 5$$

$$\begin{aligned} R_Y(\tau) &= E[(2 + 3X(t))(2 + 3X(t + \tau))] \\ &= E[4 + 6X(t + \tau) + 6X(t) + 9X(t)X(t + \tau)] \\ &= 4 + 6 + 6 + 9R_X(\tau) \\ &= 16 + 9R_X(\tau) \end{aligned}$$

自相关函数只与时间间隔 τ 有关

$\therefore Y(t)$ 为宽平稳过程

$$(2). R_Y(0) = 16 + 9R_X(0)$$

$$\because R_X(0) - 1 = 2 \Rightarrow R_X(0) = 3$$

$$\therefore R_Y(0) = 16 + 9 \times 3 = 43$$

$$(3). D[Y(t)] = D[2 + 3X(t)] = 9D(X) = 18$$

$$p(x) = \frac{1}{2\sqrt{\pi}} \exp\left[-\frac{(x-1)^2}{4}\right]$$

$$p(y) = \frac{1}{6\sqrt{\pi}} \cdot \exp\left[-\frac{(y-5)^2}{36}\right]$$

2-15. RC低通滤波器的传输特性为:

$$H(\omega) = \frac{1}{1+j\omega RC}$$

功率谱密度: $P_o(\omega) = |H(\omega)|^2 P_i(\omega)$

$$= \frac{n_0}{2} \cdot \frac{1}{1+(\omega RC)^2}$$

$$= \frac{n_0}{2} \cdot \frac{1}{1+(2\pi f RC)^2}$$

相关函数:

$$\therefore \frac{2\alpha}{\alpha^2 + \omega^2} \longleftrightarrow e^{-\alpha|t|}$$

$$\therefore \frac{n_0}{4RC} \cdot \frac{2 \cdot \frac{1}{RC}}{\omega^2 + (\frac{1}{RC})^2} \longleftrightarrow \frac{n_0}{4RC} e^{-\frac{|t|}{RC}}$$

$$\text{相关函数 } R_o(\tau) = \frac{n_0}{4RC} \cdot e^{-\frac{|\tau|}{RC}}$$

2-19.

$$(1) X(t) = A \sin(2\pi f_c t + \theta_0)$$

$$\text{相关函数: } R_x(\tau) = \int_{-\frac{T}{2}}^{\frac{T}{2}} X(t) X(t+\tau) dt$$

$$= \int_{-\frac{T}{2}}^{\frac{T}{2}} A^2 \sin(2\pi f_c t + \theta_0) \sin(2\pi f_c (t+\tau) + \theta_0) dt$$

$$= \frac{A^2}{2} \cos 2\pi f_c \tau$$

$$\text{功率谱密度: } P(f) = \int_{-\infty}^{\infty} R_x(\tau) e^{-j2\pi f \tau} d\tau = \frac{A^2}{4} [\delta(f+f_c) + \delta(f-f_c)]$$

$$\text{平均功率: } P = R_x(0) = \frac{A^2}{2}$$

(2). 相关函数:

$$R_x(t_1, t_2) = E[X(t_1) X(t_2)]$$

$$= E[A^2 \sin(2\pi f_c t_1 + \theta) \sin(2\pi f_c t_2 + \theta)]$$

$$= \frac{A^2}{2} E[\cos 2\pi f_c (t_2 - t_1) - \cos(2\pi f_c t_1 + 2\pi f_c t_2 + 2\theta)]$$

$\therefore \theta$ 在 $[-\pi, \pi]$ 上均匀分布

$$\therefore R_x(t_1, t_2) = \frac{A^2}{2} \cos[2\pi f_c (t_2 - t_1)]$$

$$\text{令 } t_2 - t_1 = \tau$$

$$\therefore R_{x_1}(\tau) = \frac{A^2}{2} \cos 2\pi f_c \tau$$

$$\text{功率谱密度: } P(f) = \frac{A^2}{4} [\delta(f+f_c) + \delta(f-f_c)]$$

$$\text{平均功率: } P = R_{x_1}(0) = \frac{A^2}{2}$$

(3). 相关函数:

$$R_{x_2}(\tau) = E[X_2(t) X_2(t+\tau)]$$

$$= E[M(t) \sin(2\pi f_c t + \theta) M(t+\tau) \sin(2\pi f_c (t+\tau) + \theta)]$$

$$= R_M(\tau) E[\sin(2\pi f_c t + \theta) \sin(2\pi f_c t + 2\pi f_c \tau + \theta)]$$

$$= \frac{1}{2} R_M(\tau) \cos 2\pi f_c \tau$$

功率谱密度:

$$P(f) = \int_{-\infty}^{+\infty} R_x(\tau) e^{-j2\pi f\tau} d\tau$$

$$= \int_{-\infty}^{+\infty} \frac{1}{2} R_m(\tau) \cos 2\pi f\tau d\tau$$

$$= \frac{1}{4} [P_m(f+f_c) + P_m(f-f_c)]$$

平均功率: $P = R(0) = \frac{1}{2} R_m(0)$

(4). $R_{x_3}(\tau) = E[X_3(t) X_3(t+\tau)]$

$$= E[M(t) \sin(2\pi f_c t + \theta_0) M(t+\tau) \sin(2\pi f_c(t+\tau) + \theta_0)]$$

$$= \frac{1}{2} R_m(\tau) [\cos(2\pi f_c \tau) - \cos(4\pi f_c t + 2\pi f_c \tau + 2\theta_0)]$$

功率谱密度:

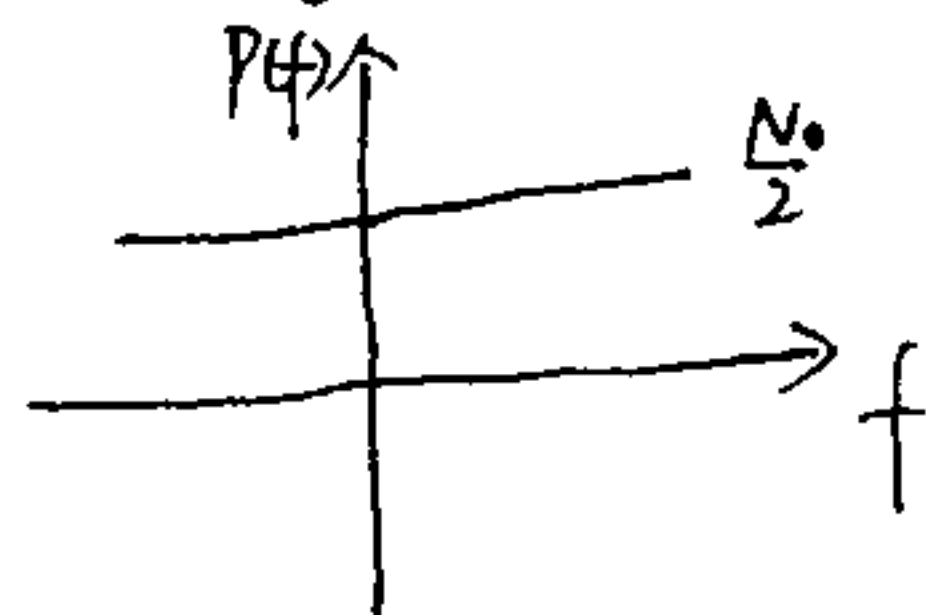
$$R_{x_3}(\tau) = \frac{1}{2} R_m(\tau) \cos 2\pi f_c \tau$$

$$P_{x_3}(f) = \frac{1}{4} [P_m(f+f_c) + P_m(f-f_c)]$$

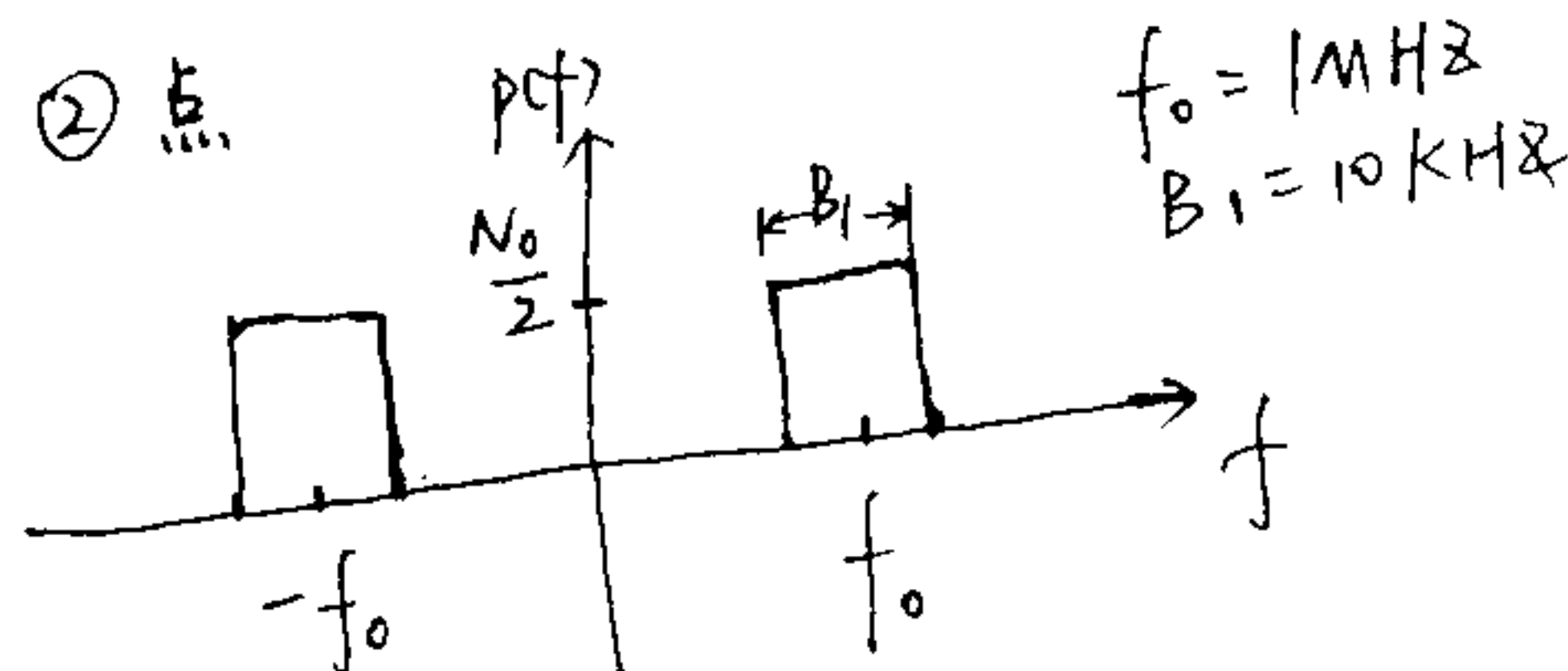
平均功率: $P = \bar{R}(0) = \frac{1}{2} R_m(0)$

2-25.

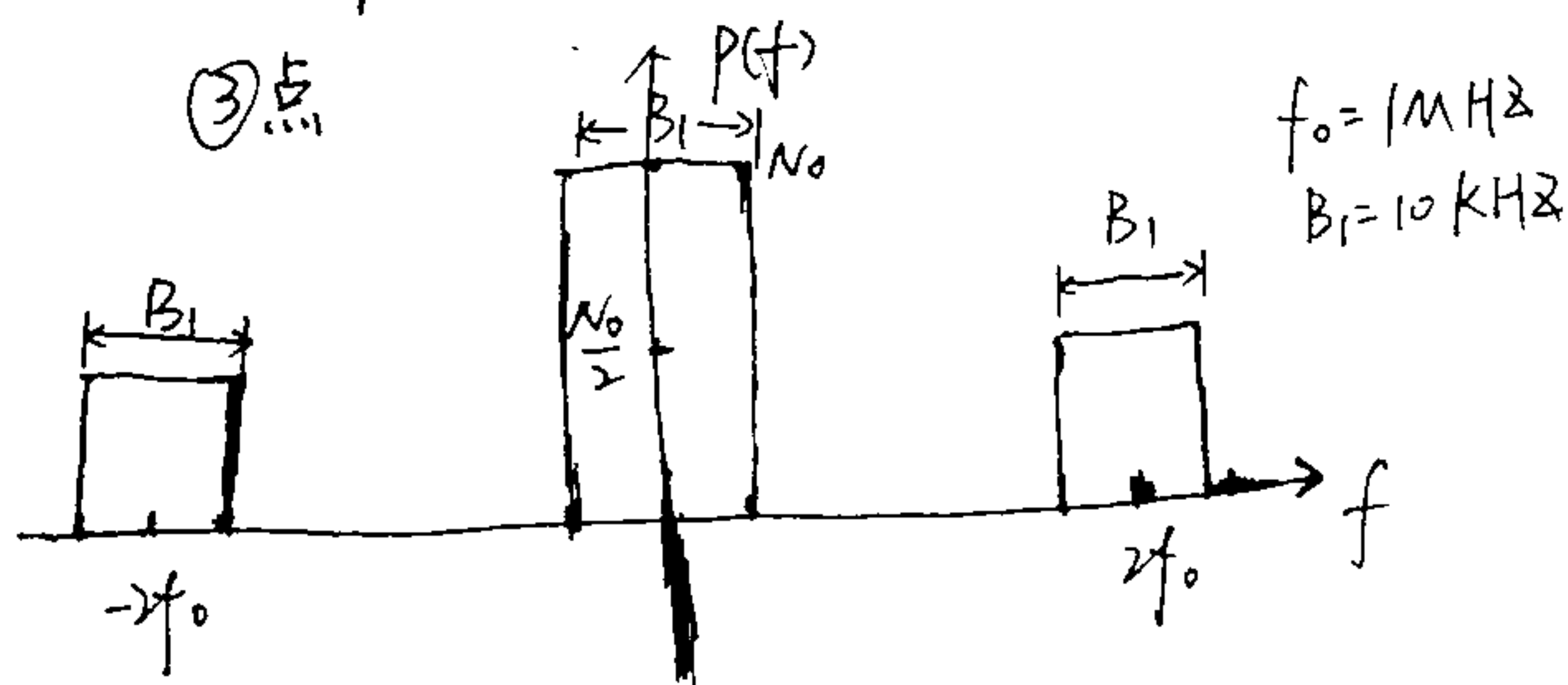
(1). ①点:



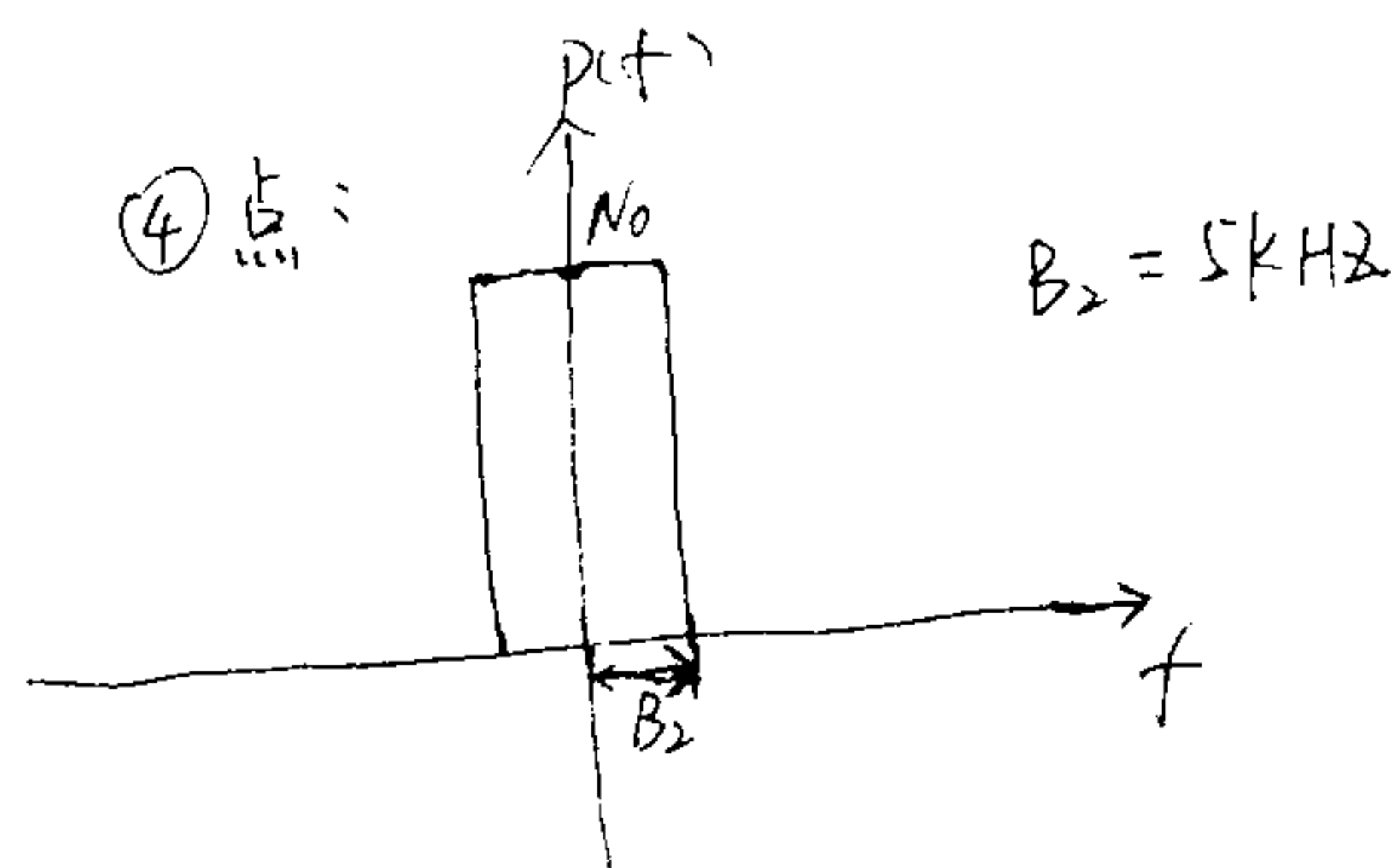
②点



③点



④点:



(2). ②点的噪声功率: $P_2 = \frac{N_0}{2} \cdot B_1 \times 2 = N_0 B_1 = 2 \times 10^{-10} \times 10 \times 10^3 = 2 \times 10^{-6} \text{ W}$

④点的噪声功率: $P_4 = N_0 \cdot 2 \cdot B_2 = 2 N_0 B_2 = 2 \times 2 \times 10^{-10} \times 5 \times 10^3 = 2 \times 10^{-6} \text{ W}$

(3). ①点的信号表示: $s_1(t) = m(t) \cos 2\pi f_0 t$

②点的信号表示: $s_2(t) = m(t) \cos 2\pi f_0 t$

③点的信号表示: $s_3(t) = 2m(t) (\cos 2\pi f_0 t)^2 = 2m(t) [\frac{1}{2} + \frac{1}{2} \cos 4\pi f_0 t]$

④点的信号表示: $s_4(t) = m(t)$

∴ ②点的信号功率: $P_2 = E[s_2^2(t)] = E[m^2(t) \cos^2(2\pi f_0 t)] = \frac{1}{2} E[m^2(t)] = \frac{1}{2} \text{ mW}$

④点的信号功率: $P_4 = E[s_4^2(t)] = E[m^2(t)] = 1 \text{ mW}$